Pathway and Network Analysis of Differential Therapeutic Response in Septic Shock Patients

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Septic Shock, Therapy Response, and Data Format

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Critical Care

RESEARCH Open Access

Identification of a transcriptome profile associated with improvement of organ function in septic shock patients after early supportive therapy

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Early **transcriptomic changes** in septic shock patients are associated with mortality

Septic shock patients classified as **non-responders (NR)** or **responders (R)** to hemodynamic therapy

R patients exhibit early **improvement in organ function**, associated with **lower mortality**

Aim to identify **gene expression profiles distinguishing R & NR patients**

- **14 NR** and **17 R** patients
- Whole blood samples taken at:
  - **T1 (within 16 hours of ICU admission)**
  - **T2 (48 hours later)**
- **62** samples sequenced and annotated with **58,096** genes

Series GSE110487 Query DataSets for GSE110487

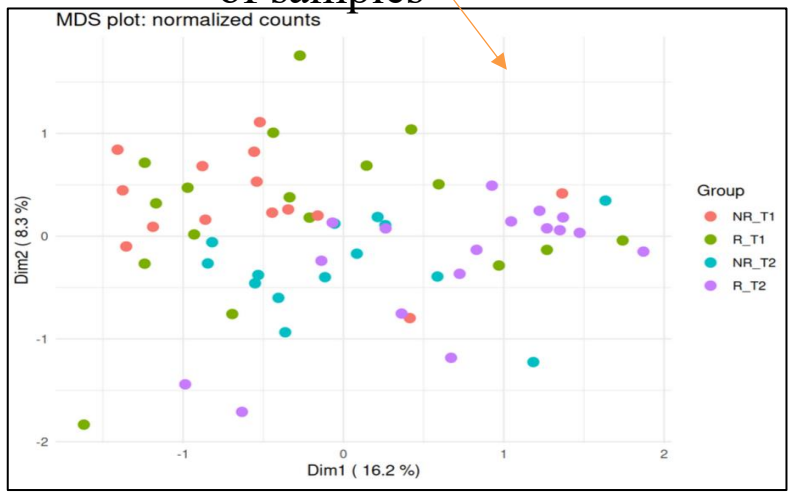
Status	Public on Dec 03, 2018
Title	Whole blood transcriptome analysis of Septic shock patients according to early therapy response
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing

Differential Expression and Thresholded Enrichment Analysis

Download 58,096 gene by 62 sample data file from GEO

Map HGNC IDs to ENSG IDs and **clean** unmapped and low expression genes, leaving **13,673 genes**

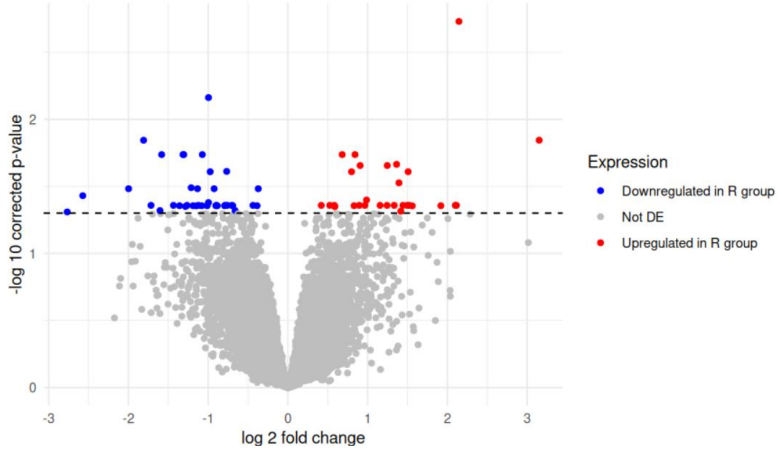
Normalize read counts and visualize distribution of samples

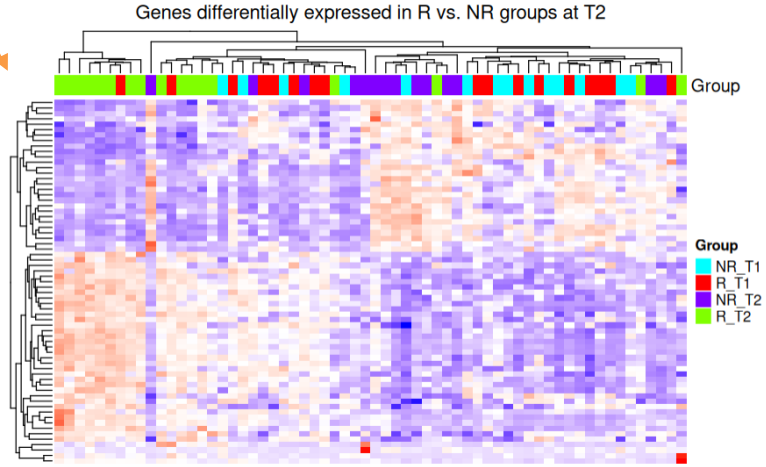


Analyze differential expression between groups at each time point, and between time points for each group

Comparison	# DE genes (corrected p < 0.05)
R vs. NR at T1	0
R vs. NR at T2	67
T1 vs. T2 in NR	846
T1 vs. T2 in R	1613

Extract and visualize **DE genes for R vs. NR at T2**





Thresholded ORA of DE genes

Term name	Term ID	p_{adj}	$-\log_{10}(p_{adj})$
regulation of antigen receptor-mediated signaling pat...	REGULATION O...	3.816×10^{-3}	
regulation of immune system process	REGULATION O...	4.986×10^{-3}	
regulation of cellular process	REGULATION O...	1.523×10^{-2}	
regulation of biological process	REGULATION O...	1.790×10^{-2}	
negative regulation of cell migration	NEGATIVE REGU...	2.291×10^{-2}	
regulation of immune response	REGULATION O...	2.472×10^{-2}	
negative regulation of cell motility	NEGATIVE REGU...	3.022×10^{-2}	
negative regulation of locomotion	NEGATIVE REGU...	3.413×10^{-2}	
regulation of T cell receptor signaling pathway	REGULATION O...	3.911×10^{-2}	

Non-thresholded Enrichment Analysis: Methods

Terms from the 2026/03/02 release of the **Bader lab genesets** downloaded and cleaned

Index of /EM_Genesets/current_release/Human/symbol

Name	Last modified	Size	Description
Parent Directory		-	
symbol_translation_summary.log	2026-03-03 16:56	489K	
Human_GOBP_AllPathways_noPFOCR_no_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	13M	
Human_GOBP_AllPathways_noPFOCR_with_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	14M	
Human_GOBP_AllPathways_withPFOCR_no_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	15M	
Human_GOBP_AllPathways_withPFOCR_with_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	16M	
Human_GO_AllPathways_noPFOCR_no_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	18M	
Human_GO_AllPathways_noPFOCR_with_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	19M	
Human_GO_AllPathways_withPFOCR_no_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	20M	
Human_GO_AllPathways_withPFOCR_with_GO_iea_March_02_2026_symbol.gmt	2026-03-03 16:56	21M	
Human_AllPathways_withPFOCR_March_02_2026_symbol.gmt	2026-03-03 16:56	9.5M	
Human_AllPathways_noPFOCR_March_02_2026_symbol.gmt	2026-03-03 16:56	7.5M	
Pathways/	2026-03-03 16:53	-	
Misc/	2026-03-03 09:36	-	
DrugTargets/	2026-03-03 09:36	-	
DiseasePhenotypes/	2026-03-03 09:36	-	
TranscriptionFactors/	2026-03-03 09:36	-	
miRs/	2026-03-03 09:36	-	
GO/	2026-03-03 09:36	-	

Merico, Daniele, Ruth Isserlin, Oliver Stueker, Andrew Emili, and Gary D. Bader. 2010. "Enrichment Map: A Network-Based Method for Gene-Set Enrichment Visualization and Interpretation." *PLoS ONE* 5 (11): e13984. https://doi.org/10.1371/journal.pone.0013984.

```r
# read and clean the downloaded gmt file so that it can be handled by gsea
cleaned_gmt_file = paste0(data_dir, "/cleaned_", BiocGenerics::basename(dest_gmt_file))
if (!file.exists(cleaned_gmt_file)) {
  gmt_lines <- readLines(dest_gmt_file, warn = FALSE)
  # remove non-ASCII characters
  clean_lines <- stringi::stri_trans_general(gmt_lines, id = "Any-Latin; Latin-ASCII")
  # remove duplicate terms
  unique_clean_lines <- clean_lines[!data.table::duplicated(clean_lines)]
  writeLines(unique_clean_lines, cleaned_gmt_file)
}
```

Genes ranked according to **significance** and **direction** of their change in expression in R vs. NR groups at T2

```r
#get all genes, calculate ranks, and arrange by rank
ranked_genes <- as.data.frame(response_T2) %>%
  dplyr::mutate(rank = -log(padj, base = 10) * sign(log2FoldChange)) %>%
  dplyr::arrange(desc(rank)) %>%
  tibble::rownames_to_column("geneid") %>%
  dplyr::select(c(geneid, rank))

#write the ranked gene list to a file to input into gsea algorithm
rank_file <- file.path(data_dir, "ranked_genes.rnk")
write.table(ranked_genes, rank_file, quote = FALSE, sep = "\t", row.names = FALSE)
```

GSEA 4.4.0 (Gene set enrichment analysis)

```r
gsea_jar <- file.path(data_dir, "GSEA_4.4.0", "gsea-cli.sh")
analysis_name <- "T2_R_vs_NR"

gsea_cmd <- paste("", gsea_jar,
                  "GSEAPreRanked -gmx", cleaned_gmt_file,
                  "-rnk" , rank_file,
                  "-collapse false -nperm 1000 -scoring_scheme weighted",
                  "-rpt_label", analysis_name,
                  "-plot_top_x 20 -rnd_seed 6789 -set_max", max_genes,
                  "-set_min", min_genes, "-zip_report false",
                  "-out", data_dir,
                  "> gsea_output.txt",sep = " ")

gsea_run <- FALSE
if(length(list.files(data_dir, pattern = "^T2_R_vs_NR\\.GseaPreranked.")) == 0 & !gsea_run) {
  suppressWarnings(system(gsea_cmd))
  gsea_run <- TRUE
}
```

Subramanian, Aravind, Pablo Tamayo, Vamsi K. Mootha, et al. 2005. "Gene Set Enrichment Analysis: A Knowledge-Based Approach for Interpreting Genome-Wide Expression Profiles." *Proceedings of the National Academy of Sciences* 102 (43): 15545–50. https://doi.org/10.1073/pnas.0506580102.

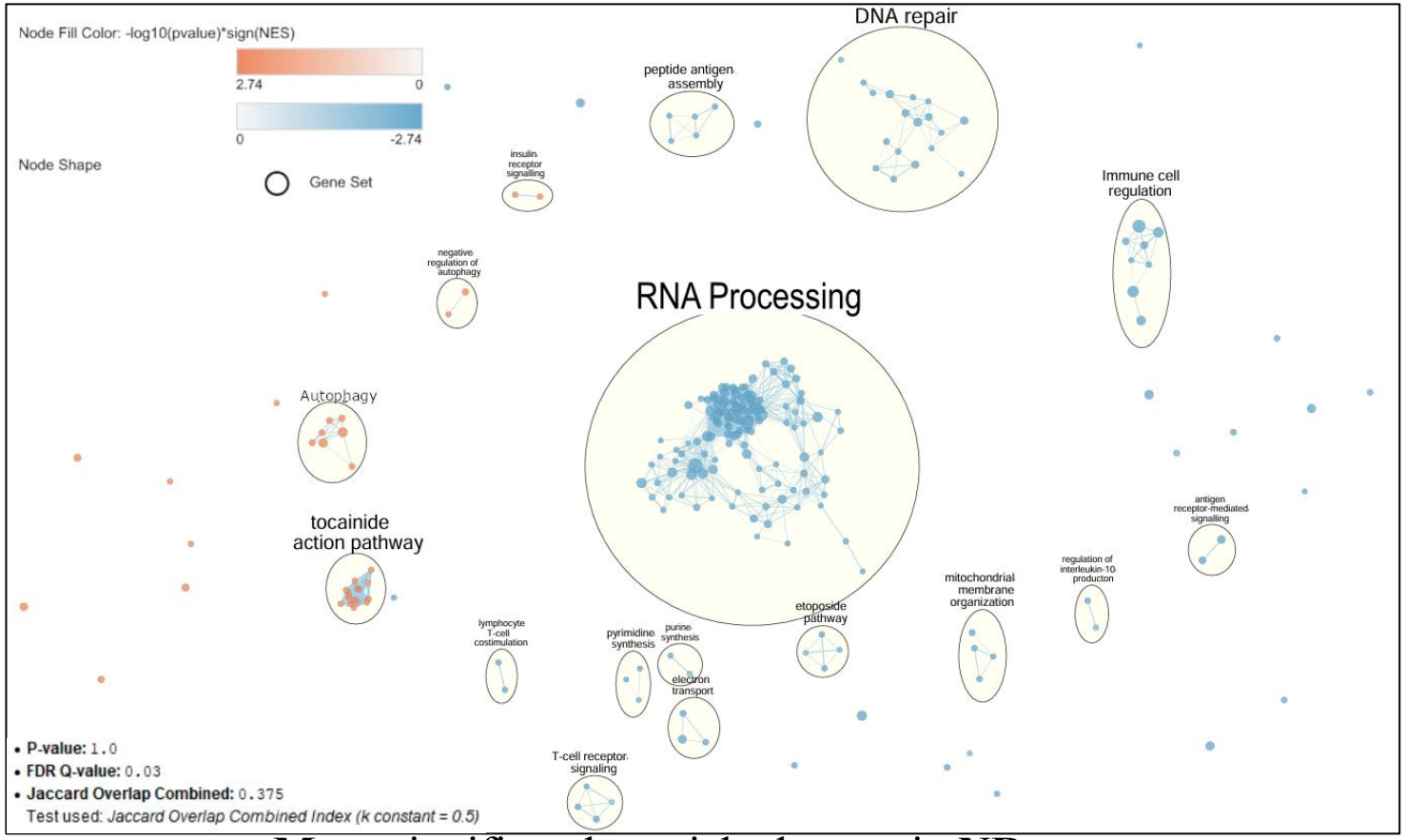
Non-thresholded Enrichment Analysis: Results

*Top enriched genesets in **R** group:*

NAME	SIZE	NES	FDR q-val
INSULIN IGF PATHWAY-PROTEIN KINASE B SIGNALING CASCADE%PANTHER PATHWAY%P00033	18	2.259356	0.003861247
NEGATIVE REGULATION OF MACROAUTOPHAGY%GOBP%GO:0016242	27	2.215916	0.003814713
AUTOPHAGY OF MITOCHONDRION%GOBP%GO:0000422	59	2.135581	0.012276860
TBC RABGAPS%REACTOME%R-HSA-8854214.2	42	2.130614	0.010157198
MITOPHAGY%GOBP%GO:0000423	44	2.126038	0.008884385

*Top enriched genesets in **NR** group:*

NAME	SIZE	NES	FDR q-val
RRNA PROCESSING%REACTOME%R-HSA-72312.5	205	-2.989837	0
RRNA PROCESSING IN THE NUCLEUS AND CYTOSOL%REACTOME%R-HSA-8868773.5	188	-2.948194	0
MAJOR PATHWAY OF RRNA PROCESSING IN THE NUCLEOLUS AND CYTOSOL%REACTOME%R-HSA-6791226.5	178	-2.909105	0
EUKARYOTIC TRANSLATION ELONGATION%REACTOME DATABASE ID RELEASE 95%156842	89	-2.871037	0
PROTEIN SYNTHESIS: PHENYLALANINE%PATHWHIZ%PW112934	80	-2.850532	0



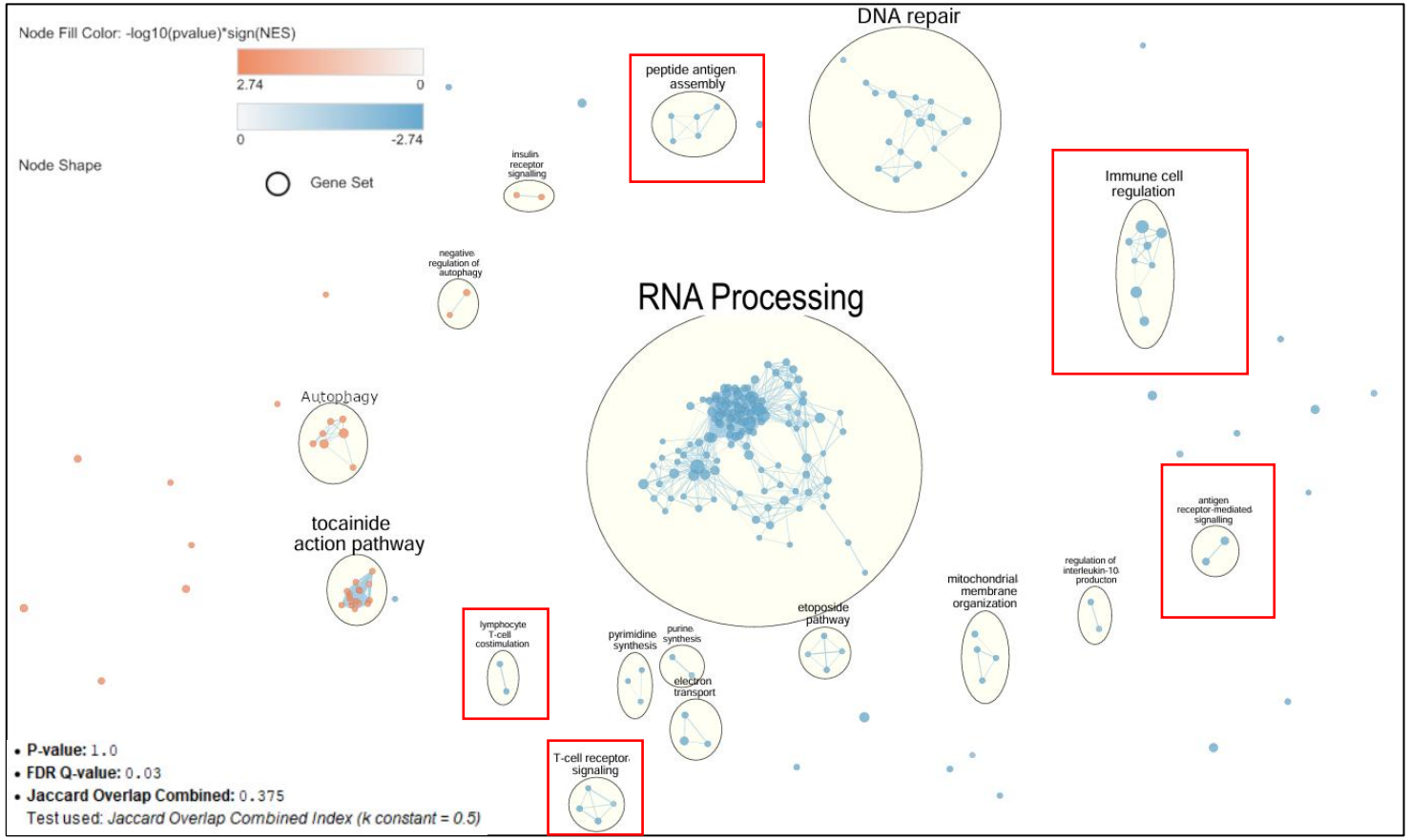
- **More** significantly enriched terms in **NR** group
  - RNA processing
  - Immune cell mediated pathways
  - DNA repair
- **Less** significantly enriched terms in **R** group
  - Autophagy
  - Insulin receptor signalling

Immune Cell Mediated Pathway Enrichment

***Thresholded** ORA hits*

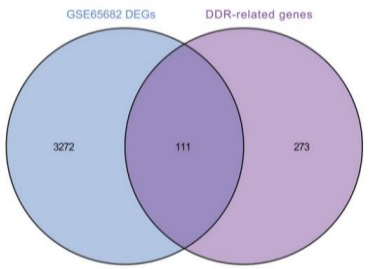
Term name
regulation of antigen receptor-mediated signaling pat...
regulation of immune system process
regulation of cellular process
regulation of biological process
negative regulation of cell migration
regulation of immune response
negative regulation of cell motility
negative regulation of locomotion
regulation of T cell receptor signaling pathway

- Terms related to **immune cell pathways** were significant in both analyses
- Terms related to cell migration were not highlighted in the non-thresholded analysis
- Non-thresholded analysis identified many significant terms not identified in the thresholded analysis



"Interestingly, a comparison of the biological networks between Rs and NRs highlighted that R patients specifically modulate genes involved in T-cell activation but that NR patients do not, as observed in the trend analysis. In detail, ICOS and LCK, two important co-stimulatory molecules of T-cell activation [25, 26], and CD247, the major signal transduction element in the T-cell antigen receptor complex, were upregulated in Rs at T2. Modulation of these genes has been observed in relation to survival in previous articles [23, 27]." (Barcella et al. 2018)

Other Enriched Terms



- Recent study reveals **DNA damage repair (DDR) genes as sepsis biomarkers**
- DNA repair gene enrichment in **NR group** support a correlation between **negative response to therapy** and **higher sepsis biomarker expression**

  "However, studies elucidating the DDR gene's involvement in sepsis remain scarce. Therefore, it is imperative to uncover the roles of DDR in sepsis." (Gu et al. 2025)

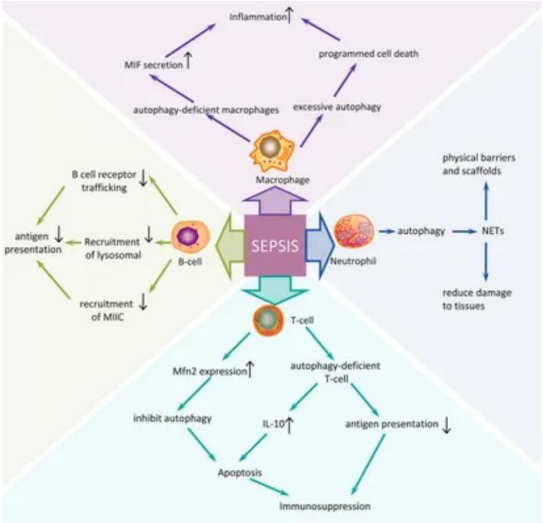
Gu, Jin, Dong-Fang Wang, and Jian-Ying Lou. 2025. "Identification of DNA Damage Repair-Related Genes in Sepsis Using Bioinformatics and Machine Learning: An Observational Study." *Medicine* 104 (5): e41354. https://doi.org/10.1097/md.0000000000041354.

- Enrichment of nondescript terms related to RNA processing in the **NR group** may reflect **higher expression** among therapy non-responders of **common gene classes upregulated by septic pathways**

  "At the onset, sepsis generally originates from identifying diverse microbial products resulting from infection and endogenous danger molecular drivers engaging with specific cell-surface receptors on host cells, initiating downstream signaling cascades. Subsequently, the activation of these diverse signaling pathways ultimately results in **upregulating several common gene classes** associated with inflammation, adaptive immunity, and cellular metabolism." (Bushra et al. 2024)

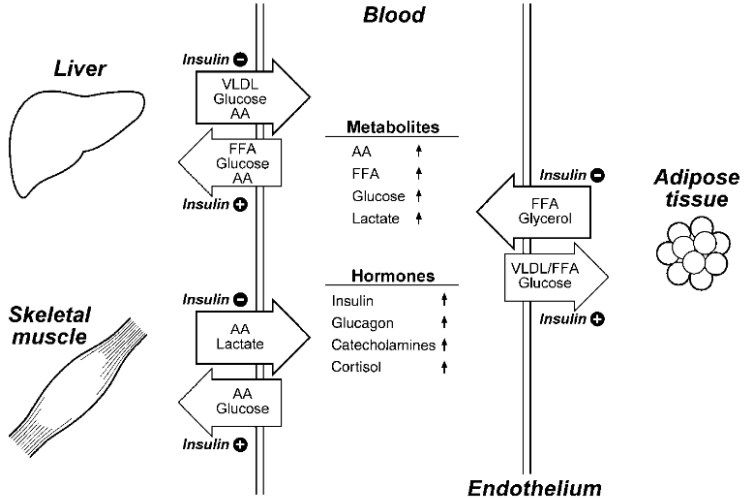
Bushra, Shaik Iqbal Ahmed, Safia Begum, et al. 2024. "Molecular Basis of Sepsis: A New Insight into the Role of Mitochondrial DNA as a Damage-Associated Molecular Pattern." *Mitochondrion* 79 (November): 101967. https://doi.org/10.1016/j.mito.2024.101967.

- Autophagy **dysregulated** by septic pathways, causing **tissue damage**
- Enrichment of autophagy related genes in **R group** may reflect **regulation of dysregulated autophagy**
- May contribute to organ function improvement



Yin, Xin, Huang Xin, Shuai Mao, Guangping Wu, and Liheng Guo. 2019. "The Role of Autophagy in Sepsis: Protection and Injury to Organs." *Frontiers in Physiology* 10 (August). https://doi.org/10.3389/fphys.2019.01071.

- Sepsis **upsets insulin balance** and promotes insulin resistance
- Enrichment of insulin pathway genes in **R group** may reflect restoration of balance



Andersen, S. K., Gjedsted, J., Christiansen, C., & Tønnesen, E. (2003). The roles of insulin and hyperglycemia in sepsis pathogenesis. *Journal of Leukocyte Biology*, *75*(3), 413–421. https://doi.org/10.1189/jlb.0503195